



# MACHINE LEARNING

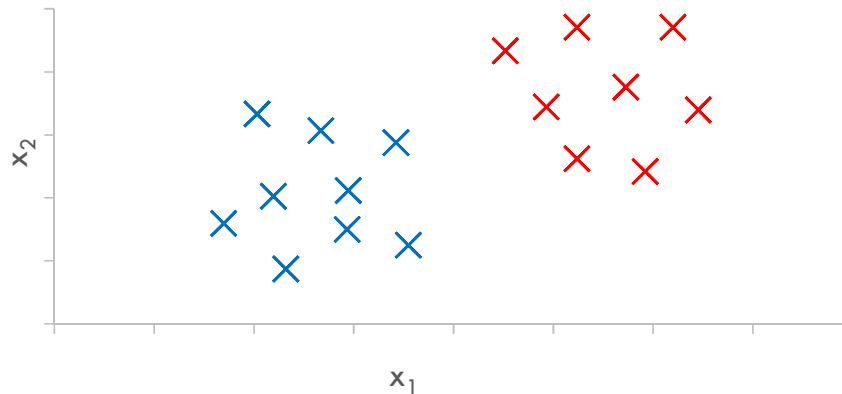
## LESSON 06: Clustering

Prepared by Dr. Dona Valy

Inspired by and adapted from the Coursera Machine Learning course by Andrew Ng  
([coursera.org/learn/machine-learning](https://coursera.org/learn/machine-learning))

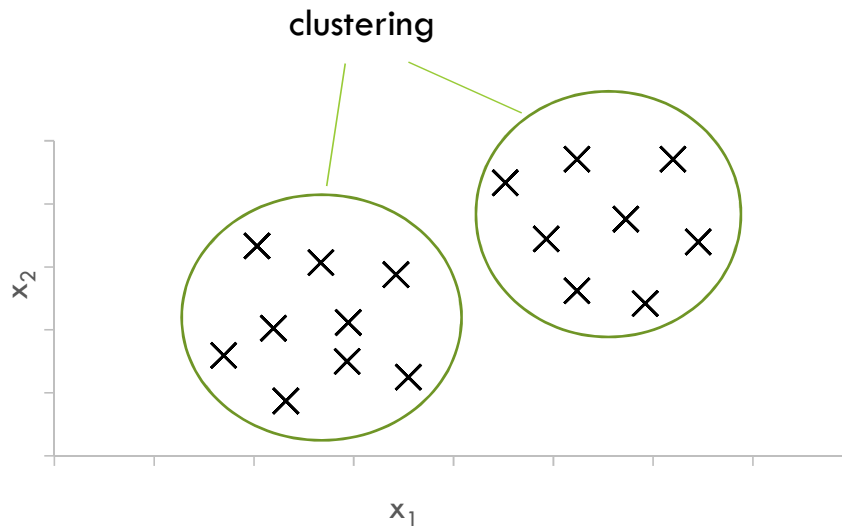
# SUPERVISED LEARNING

- Training set:  $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$



# UNSUPERVISED LEARNING

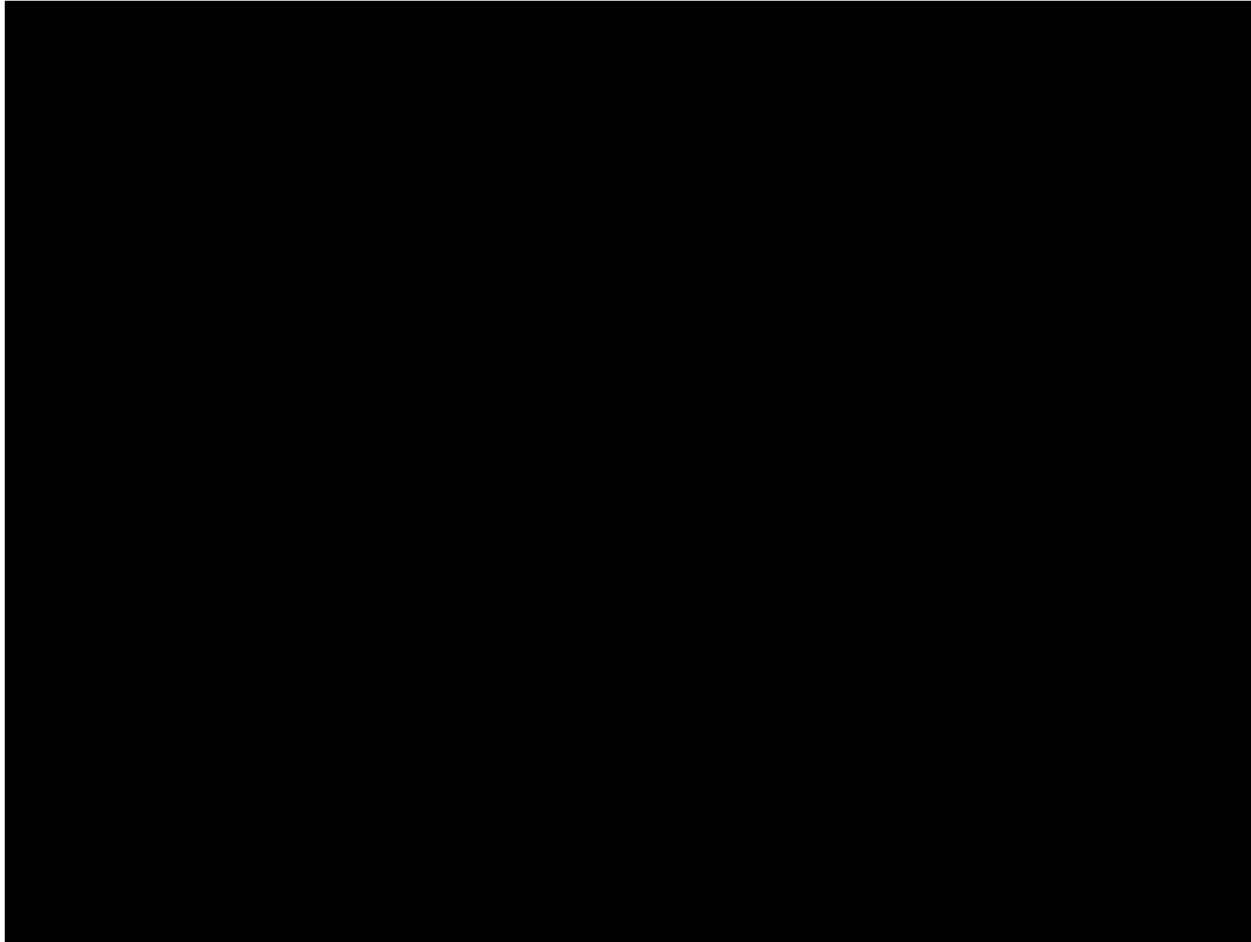
- Training set:  $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$



Some examples:

- Social network analysis
- Market segmentation
- Computing clusters
- Astronomical data analysis

# K-MEANS



Credit: <https://www.youtube.com/watch?v=5I3Ei69I40s>

# K-MEANS

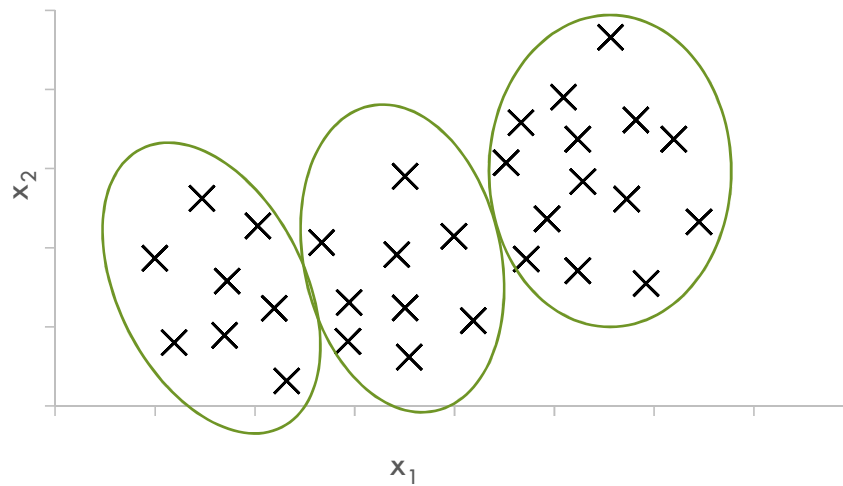
- Inputs:
  - $K$  number of clusters
  - Training set  $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}, x^{(i)} \in \mathbb{R}^n$

# K-MEANS

- Randomly initialize  $K$  cluster centroids  $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$
- REPEAT
  - Assign data points to cluster:  
for  $i = 1$  to  $m$   
$$c^{(i)} := \operatorname{argmin}_k \|x^{(i)} - \mu_k\|^2 \text{ (index of the cluster centroid closest to } x^{(i)})$$
  - Move centroids  
for  $k = 1$  to  $K$   
$$\mu_k := \text{average of points assigned to cluster } k$$

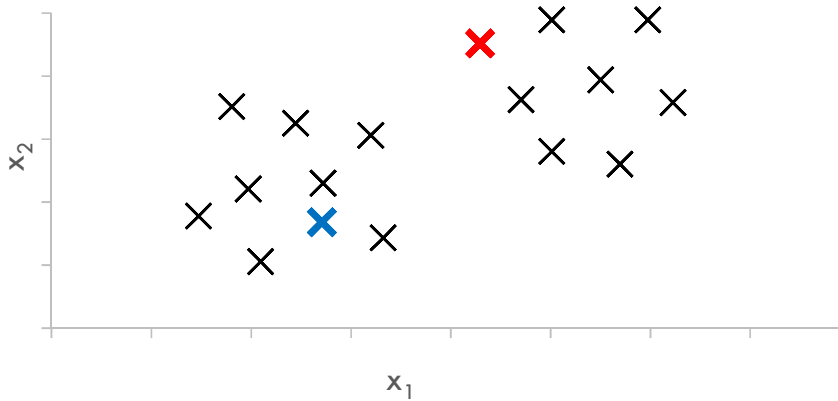
# K-MEANS

- Non-separated clusters



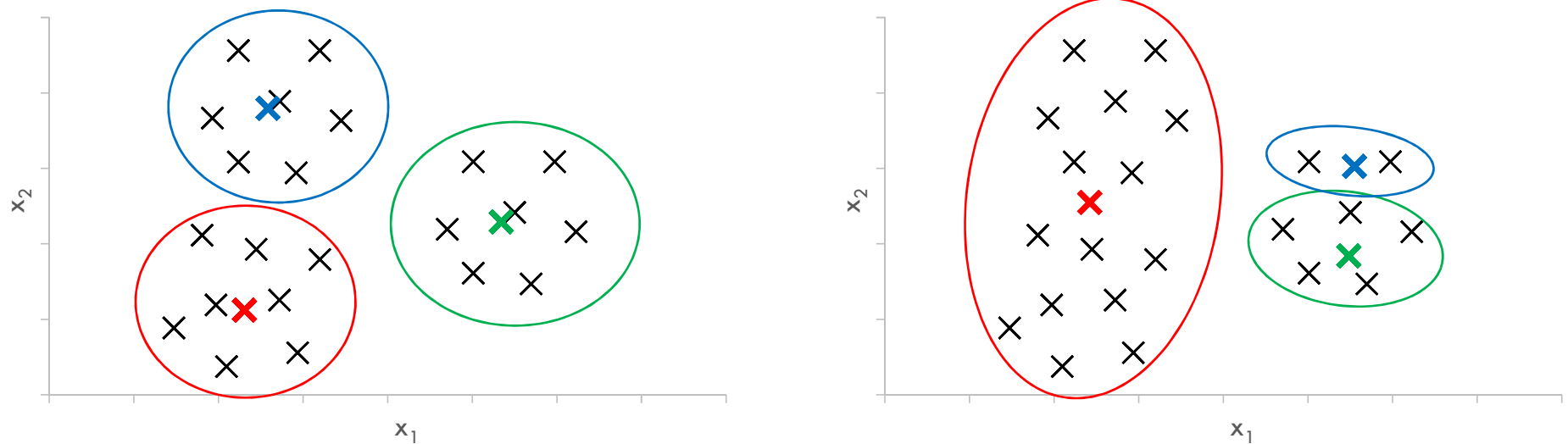
# K-MEANS

- Random initialization of  $K$  cluster centroids  $\mu_1, \mu_2, \dots, \mu_K$ 
  - Should have  $K < m$
  - Randomly pick  $K$  training examples as centroids



# K-MEANS

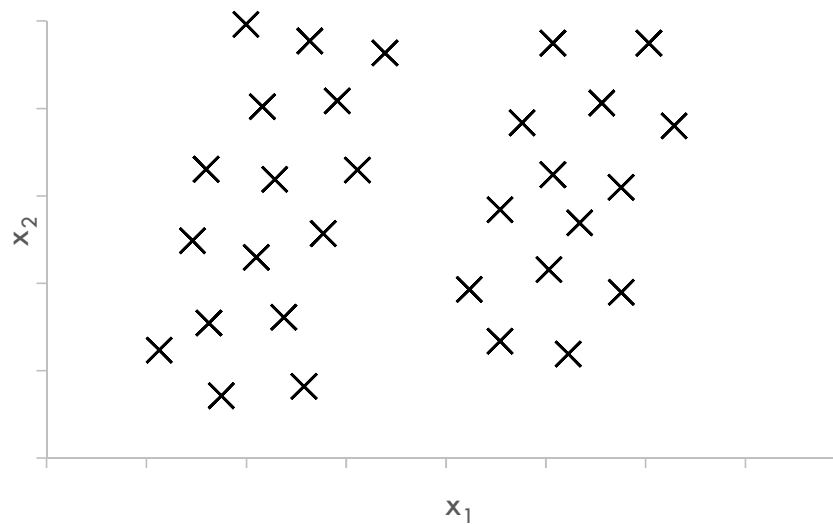
- Local optima



Apply K-means clustering multiple times, each time compute  $J = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$  where  $c^{(i)}$  the index of the centroid that  $x^{(i)}$  belongs to,  $1 \leq c^{(i)} \leq K$   
 $\Rightarrow$  Choose the clustering that gives the lowest  $J$

# NUMBER OF CLUSTERS

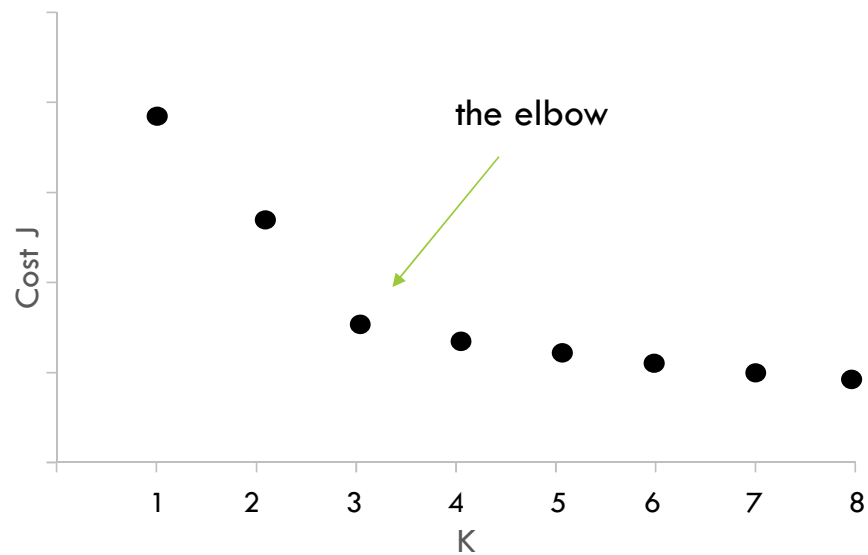
- What is the right number of  $K$ ?



The number of clusters  $K$  can be ambiguous!

# NUMBER OF CLUSTERS

- Elbow method:



## EXERCISE 6.1

Given a dataset 'data\_6\_1.csv' containing 200 data points, apply K-means to cluster them into  $K = 5$  clusters. Visualize the result.

**Note:** you should run the algorithm multiple times and select the result which corresponds to the lowest cost  $J$ .

## EXERCISE 6.2: COLOR QUANTIZATION

Color quantization finds a small number of representative colors within a given picture. Each pixel yields one 3-dimensional vector in the RGB color space. Using K-means we can cluster all the pixels of an image into K clusters and assign each pixel the color represented by its nearest cluster center. Thereby, an image containing millions of colors can be compressed to an image containing K different colors only.

- Load an image of your choice, treat each pixel as an individual 3-dimensional data point and cluster them into K clusters
- Assign each pixel the color value of its nearest cluster center
- Visualize the result

# OTHER CLUSTERING ALGORITHMS

- K-medians
- Hierarchical clustering
- DBSCAN
- ...etc

